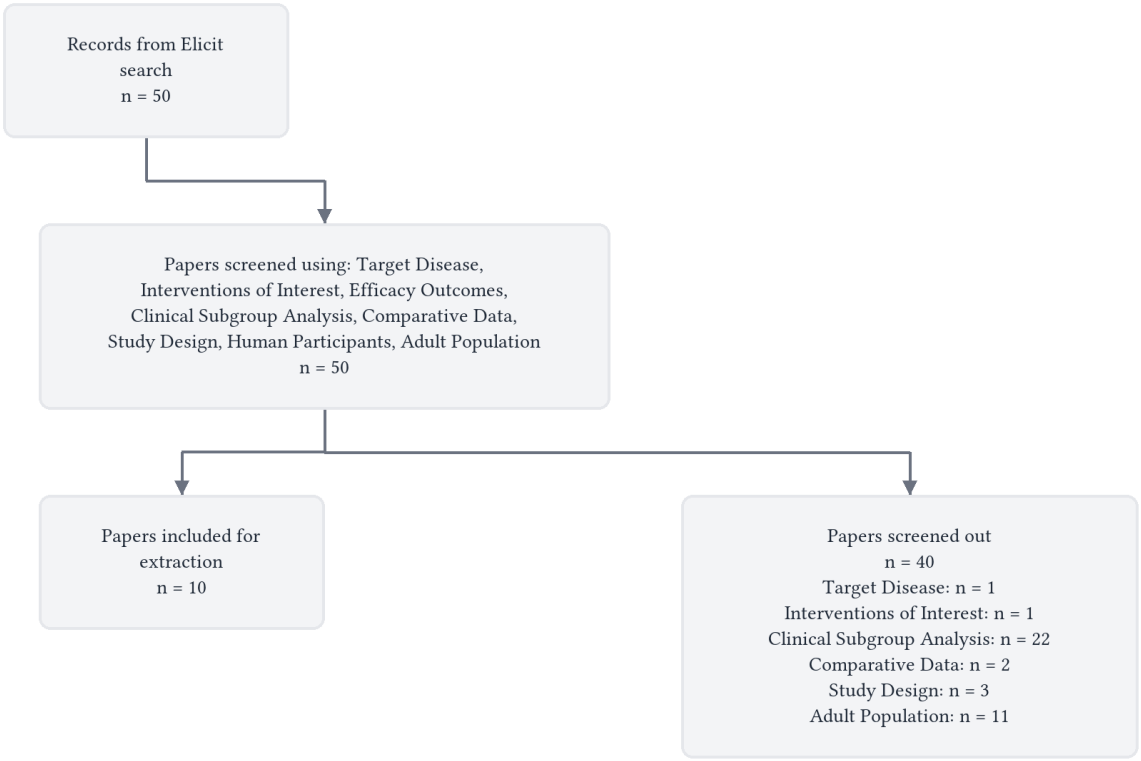
# ¿Cómo varía la eficacia de belimumab versus anifrolumab según el perfil clínico de los pacientes (por ejemplo, presencia de nefritis lúpica, manifestaciones cutáneas, o actividad serológica)?

Anifrolumab es superior a belimumab en las manifestaciones cutáneas y en pacientes con alta actividad serológica e interferón tipo I elevado, belimumab es el único agente con datos controlados en nefritis lúpica, y en la actividad sistémica global ambos fármacos son equivalentes, por lo que la selección debe basarse en el fenotipo clínico dominante del paciente.


## Abstract

The comparative efficacy of belimumab and anifrolumab in systemic lupus erythematosus varies substantially by clinical domain, and no direct head-to-head trial data exist to resolve this question definitively. For overall disease activity measured by SRI-4, indirect comparative analyses reach contradictory conclusions—one finding belimumab superior (OR 1.99, 95% CI 1.25–3.16) [1], another finding anifrolumab superior (OR 2.61, 95% CI 1.22–5.58) [2], and a network meta-analysis finding no significant difference between the two [3] —discordance that most plausibly reflects residual population heterogeneity between the TULIP and BLISS trial programmes rather than a true pharmacological difference. In the cutaneous domain, the evidence consistently and substantially favors anifrolumab: controlled trial data show a 21-percentage-point advantage over placebo in CLASI-A response at week 12 among patients with active skin disease [4], a benefit that is concentrated in IFNGS-high patients (22.6 pp, p=0.002) and absent in IFNGS-low patients (6.7 pp, p=0.692) [4], while real-world series confirm meaningful CLASI reductions even in patients who had previously failed belimumab [5, 6]. For serologically active disease, anifrolumab's BICLA advantage over placebo was 23.1 pp in patients with abnormal baseline serology but only 8.5 pp (non-significant) in those with normal serology [4], indicating that seropositive patients represent the subgroup with the clearest anifrolumab benefit. For lupus nephritis, the comparative question is empirically unanswerable at present: anifrolumab lacks regulatory approval for this indication and has been evaluated in only one published phase II trial [1], the indirect complete renal response comparison is statistically uninformative (OR 0.83, 95% CI 0.22–3.10) [1], and only belimumab has controlled trial data supporting renal use. Across all comparisons, the principal safety distinction is a consistently elevated herpes zoster risk with anifrolumab versus belimumab (NMA OR 4.74, 95% CI 2.00–11.21) [3], though prior vaccination may attenuate this risk in clinical practice [6]. Taken together, the available evidence supports individualized biologic selection driven by dominant clinical phenotype: anifrolumab for patients with prominent cutaneous involvement or high serological and interferon activity, and belimumab for patients with lupus nephritis or predominantly musculoskeletal disease in whom high baseline SLEDAI-2K and polyarthritis predict response [7].

## Flow Diagram




## Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "¿Cómo varía la eficacia de belimumab versus anifrolumab según el perfil clínico de los pacientes (por ejemplo, presencia de nefritis lúpica, manifestaciones cutáneas, o actividad serológica)?"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

## Screening

We screened in sources based on their abstracts that met these criteria:

- **Target Disease**: Does the study focus on patients with systemic lupus erythematosus (SLE)?
- **Interventions of Interest**: Does the study include belimumab and/or anifrolumab as an intervention?
- **Efficacy Outcomes**: Does the study report at least one efficacy outcome (e.g., disease activity scores such as SLEDAI, BILAG, SRI, BICLA, flare rates, organ-specific response rates, or serological markers)?
- **Clinical Subgroup Analysis**: Does the study report efficacy results stratified by at least one clinical subgroup or patient profile (e.g., lupus nephritis, cutaneous manifestations, serological activity, anti-dsDNA positivity, complement levels, or IFNGS status)?
- **Comparative Data**: Does the study provide a direct or indirect comparison between belimumab and anifrolumab, or does it provide subgroup-level efficacy data for at least one of these agents that could be used in a comparative analysis (including systematic reviews and meta-analyses)?

- **Study Design**: Is the study a randomized controlled trial, observational study, systematic review, or meta-analysis (i.e., not a case report, editorial, letter, or narrative review without original data)?
- **Human Participants**: Does the study involve human participants (i.e., not an in vitro or animal study)?
- **Adult Population**: Does the study include adult patients (≥18 years of age), or a mixed population where adult data can be extracted separately?

Papers that failed any criterion were automatically excluded; all other papers, including those with a Maybe on any criterion, were screened in.

At abstract screening, the number of papers excluded for each primary reason was:

- **Target Disease**: n = 1
- **Interventions of Interest**: n = 1
- **Clinical Subgroup Analysis**: n = 22
- **Comparative Data**: n = 2
- **Study Design**: n = 3
- **Adult Population**: n = 11

## Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Design**:

  Extract information about the study methodology relevant to interpreting the comparative efficacy of belimumab vs. anifrolumab, including:

- Study type (RCT, indirect treatment comparison, meta-analysis, real-world observational study, case series, etc.)
- Comparison method used (direct head-to-head, indirect comparison such as MAIC or simulated treatment comparison, single-arm real-world)
- Sample size (total and per treatment arm)
- Duration of follow-up or treatment
- Whether the analysis is based on individual patient data (IPD) or aggregate/summary-level data
- Name(s) of the source trials or cohorts (e.g., TULIP-1/2, BLISS-52/76)

- **Patient Clinical Profile**:

  Extract details about the clinical profiles of included patients, as these are the key moderating variables for the research question. Include:

- Presence and proportion of patients with lupus nephritis (LN) and, if reported, LN class or severity
- Presence and proportion of patients with cutaneous manifestations (e.g., skin rashes, discoid lupus) and their severity at baseline (e.g., CLASI score)
- Serological activity at baseline: anti-dsDNA positivity rates, hypocomplementemia (C3/C4 levels)
- Overall disease activity at baseline (e.g., SLEDAI-2K or SELENA-SLEDAI scores, proportion with moderate-to-severe disease)
- Other dominant manifestations present (e.g., polyarthritis, musculoskeletal involvement)
- Key inclusion/exclusion criteria that define which clinical profiles were studied or excluded
- Background/concomitant immunosuppressive therapy (e.g., hydroxychloroquine, azathioprine, mycophenolate, corticosteroid dose)

- **Efficacy by Profile**:

  Extract efficacy results for belimumab and/or anifrolumab broken down by the relevant clinical subgroup or manifestation domain. This is the core data needed to answer the research question. For each drug and each clinical profile (lupus nephritis, cutaneous manifestations, serological activity, overall SLE), extract:

- Outcome measure used (e.g., SRI-4, SLEDAI-2K reduction ≥4, BILAG, CLASI-A reduction, complete/partial renal response, proteinuria change, anti-dsDNA normalization, C3/C4 normalization, corticosteroid tapering)
- Observed response rates or effect sizes with confidence intervals for each drug
- Time point of assessment (e.g., 52 weeks, 12 months, 24 months)
- Whether the result is from a pre-specified subgroup analysis, post-hoc analysis, or primary endpoint
- Any statistically significant differences in efficacy within a specific clinical profile

- **Comparative Efficacy**:

  Extract the head-to-head or indirect comparison results between belimumab and anifrolumab, overall and by clinical profile. Include:

- Odds ratios, hazard ratios, risk ratios, or other comparative effect estimates with 95% confidence intervals
- Which drug showed superior efficacy and for which specific outcome or clinical domain (e.g., anifrolumab superior for skin; belimumab superior for SRI-4)
- Statistical significance and precision of the comparison
- Method used to derive the comparison (e.g., network meta-analysis, MAIC, simulated treatment comparison, indirect meta-analytic approach)
- Limitations of the comparison acknowledged by the authors (e.g., different trial populations, different inclusion criteria, wide CIs)
- Whether superiority, non-inferiority, or uncertainty was concluded for each clinical domain

- **Response Predictors**:

  Extract clinical, demographic, or biomarker factors that predict or modify treatment response to belimumab or anifrolumab, as these directly explain how efficacy varies by patient profile. Include:

- Variables identified as independent predictors of response to belimumab (e.g., baseline SLEDAI score, polyarthritis, serological activity, corticosteroid dose)
- Variables identified as independent predictors of response to anifrolumab (e.g., high interferon gene signature, cutaneous involvement, type I IFN pathway activity)
- Variables associated with treatment failure or drug discontinuation for each agent
- Statistical method used to identify predictors (e.g., univariate/multivariate logistic regression, Cox model)
- Any differential predictor patterns suggesting that a specific clinical profile benefits more from one drug over the other

- **Safety & Discontinuation**:

  Extract safety data and treatment discontinuation information relevant to clinical decision-making by patient profile, including:

- Incidence of serious adverse events for belimumab and anifrolumab
- Rates of specific adverse events relevant to each drug (e.g., herpes zoster for anifrolumab; infections for belimumab)
- Rates and reasons for treatment discontinuation for each drug
- Whether any safety signals were more pronounced in specific clinical subgroups (e.g., LN patients, heavily pre-treated patients)
- Overall drug survival rates at reported time points

- **Authors' Conclusions**:

  Extract the authors' main conclusions specifically regarding which drug (belimumab or anifrolumab) is preferred or more effective for which clinical profile of SLE patients. Include:

- Stated recommendation or conclusion per clinical domain (lupus nephritis, cutaneous SLE, serologically active SLE, high-disease-activity SLE)
- Any clinical guidance offered on patient selection for one drug over the other based on clinical profile
- Key limitations acknowledged that affect the validity of the comparative conclusions (e.g., absence of head-to-head data, heterogeneity of trial populations, small sample sizes in real-world studies)
- Identified gaps in evidence (e.g., lack of anifrolumab approval/data in LN)

# Results

## Characteristics of Included Studies

| Study | Full Text Retrieved? | Study Type | Comparison Method | Sample Size | Follow-up | Source Trials/Cohorts |
|---|---|---|---|---|---|---|
| Martínez-Martínez et al., 2026 [3] | No | Systematic review and network meta-analysis of RCTs [3] | Indirect network comparison; each biologic vs. placebo + SoC [3] | 11 trials, n=8,607 (4 anifrolumab; 7 belimumab) [3] | Not reported [3] | Not specified [3] |
| Morand et al., 2022 [8] | No | Post-hoc pooled analysis of two phase III RCTs [8] | Anifrolumab vs. placebo; no belimumab arm [8] | n=726 (anifrolumab 360; placebo 366) [8] | 52 weeks [8] | TULIP-1, TULIP-2 [8] |
| Rodríguez Tudero et al., 2025 [9] | No | Retrospective, cross-sectional, descriptive observational study [9] | Single-arm belimumab cohort [9] | n=17 total; n=10 with LN [9] | Mean 16.9 ± 10.69 months [9] | Single center; no named trial [9] |
| Garbarino et al., 2025 [6] | No | Retrospective, descriptive real-world case series [6] | Single-arm anifrolumab; no belimumab comparison [6] | n=13 [6] | Median 10 administrations; complete response at mean 5 months [6] | Single center, Hospital Clínic de Barcelona [6] |
| de la Rubia Navarro et al., 2022 [10] | Yes | Observational, retrospective, cross-sectional real-world study [10] | Single-arm belimumab cohort [10] | n=29 total; n=9 with kidney involvement [10] | Mean 4.61 years since belimumab initiation (kidney subgroup) [10] | Single tertiary hospital; no named trial [10] |
| Flouda et al., 2024 [5] | Yes | Real-world observational case series [5] | Single-arm anifrolumab; no belimumab comparison arm [5] | n=18 [5] | Mean 7.1 months [5] | Single-center; no named trial [5] |
| Iaccarino et al., 2018 [7] | No | Real-world observational analysis from 11 Italian prospective cohorts [7] | Single-treatment belimumab cohort [7] | n=188 [7] | Mean 17.5 ± 10.6 months [7] | 11 Italian prospective cohorts [7] |

| Study | Full Text Retrieved? | Study Type | Comparison Method | Sample Size | Follow-up | Source Trials/Cohorts |
|---|---|---|---|---|---|---|
| Lee & Song, 2025 [1] | No | Meta-analysis of RCTs [1] | Indirect comparison using placebo-controlled trial ORs and HRs [1] | 7 RCTs, n=4,332 [1] | Not reported [1] | Not specified [1] |
| Bruce et al., 2022 [2] | No | Indirect comparative efficacy analysis [2] | Population-adjusted simulated treatment comparison (primary); MAIC (supporting); IPD from TULIP-1/2 and aggregate data from BLISS-52/76 [2] | Not reported [2] | 52 weeks [2] | TULIP-1, TULIP-2, BLISS-52, BLISS-76 [2] |
| Vital et al., 2022 [4] | Yes | Post-hoc pooled subgroup analysis of two phase III RCTs [4] | Anifrolumab vs. placebo; no belimumab arm [4] | n=726 (anifrolumab 360; placebo 366) [4] | 52 weeks [4] | TULIP-1, TULIP-2 [4] |

The ten included sources are highly heterogeneous in design: three are indirect comparative analyses or network meta-analyses [1–3], two are post-hoc pooled analyses of the anifrolumab TULIP programme restricted to anifrolumab versus placebo [4, 8], and five are real-world single-arm or cohort studies without a head-to-head comparison [5–7, 9, 10]. No study reported a direct randomised head-to-head comparison of belimumab and anifrolumab. The total patient population across comparative analyses ranges from 726 (TULIP pooled analyses) to 8,607 (the network meta-analysis), while real-world cohorts are substantially smaller, ranging from 13 to 188 patients [6, 7]. This heterogeneity in design and evidence quality is a critical interpretive consideration throughout.

The patient populations studied also differ in important ways. The RCT-based analyses enrolled autoantibody-positive patients with moderate-to-severe SLE despite standard therapy [4], with musculoskeletal involvement present in 89–94% and mucocutaneous involvement in 86–96% of pooled TULIP patients [8]. The real-world anifrolumab series skewed toward patients with prominent or refractory cutaneous involvement [5, 6], while real-world belimumab cohorts included a substantial proportion with lupus nephritis [9, 10]. The Iaccarino cohort, the largest real-world belimumab study, found polyarthritis (45.2%) and skin rashes (25.5%) to be the most frequent indications [7]. The Vital et al. post-hoc analysis also showed that 82.6% of TULIP patients were IFNGS-high, with IFNGS-high patients having greater serological activity and higher glucocorticoid burdens at baseline compared with IFNGS-low patients [4].

## Comparative Efficacy

### Overall Disease Activity

| Outcome | Drug(s) | Effect Estimate | Method | Time Point | Notes |
|---|---|---|---|---|---|
| SRI-4 response [3] | Anifrolumab vs. placebo | OR 1.78 (95% CI 1.41–2.24) [3] | NMA [3] | Not reported [3] | No significant difference between active drugs [3] |
| SRI-4 response [3] | Belimumab vs. placebo | OR 1.62 (95% CI 1.38–1.90) [3] | NMA [3] | Not reported [3] | No significant difference between active drugs [3] |
| SRI-4 response [1] | Belimumab vs. anifrolumab (indirect) | OR 1.99 (95% CI 1.25–3.16) favoring belimumab [1] | Indirect meta-analysis [1] | Not reported [1] | Statistically significant; CI excludes 1 [1] |
| SRI-4 response [2] | Anifrolumab vs. belimumab (indirect) | OR 2.61 (95% CI 1.22–5.58) favoring anifrolumab [2] | Population-adjusted simulated treatment comparison (primary) [2] | 52 weeks [2] | Statistically significant; CI excludes 1 [2] |
| SLEDAI-2K ≥4 reduction [2] | Anifrolumab vs. belimumab (indirect) | OR 2.47 (95% CI 1.16–5.25) favoring anifrolumab [2] | Population-adjusted simulated treatment comparison [2] | 52 weeks [2] | Statistically significant [2] |
| BICLA response [3] | Anifrolumab vs. belimumab (indirect) | OR 1.56 (95% CI 1.10–2.21) favoring anifrolumab [3] | NMA [3] | Not reported [3] | Moderate certainty; relies on indirect evidence; belimumab BICLA data from post-hoc analyses [3] |
| BICLA response, total population [4] | Anifrolumab vs. placebo | Difference 16.6 pp (95% CI 9.7–23.6); nominal p<0.001 [4] | Post-hoc pooled RCT analysis [4] | 52 weeks [4] | IFNGS-high subgroup: 18.2 pp; IFNGS-low: 9.3 pp (non-significant) [4] |
| SRI-4, real-world [7] | Belimumab | 77.0% at 12 months; 68.7% at 24 months [7] | Real-world observational cohort [7] | 12 and 24 months [7] | No comparative arm [7] |
| Sustained glucocorticoid taper [1] | Anifrolumab vs. belimumab (indirect) | OR 0.31 (95% CI 0.19–0.51) favoring anifrolumab [1] | Indirect meta-analysis [1] | Not reported [1] | Statistically significant; CI excludes 1 [1] |
| Sustained glucocorticoid taper (≤7.5 mg/day, weeks 40–52) [4] | Anifrolumab vs. placebo | Difference 18.7 pp (95% CI 8.9–28.4); p<0.001 [4] | Post-hoc pooled RCT analysis [4] | Weeks 40–52 [4] | Consistent across serological subgroups [4] |

The overall efficacy landscape is markedly discordant depending on both the outcome measure and the analytical approach applied. On SRI-4, the network meta-analysis of Martínez-Martínez et al. found no statistically significant

difference between anifrolumab and belimumab versus placebo [3], yet the two indirect analyses that directly modeled one drug against the other arrived at opposite conclusions: Lee & Song reported that belimumab was superior for SRI-4 (OR 1.99) [1], whereas Bruce et al. found anifrolumab superior for both SRI-4 (OR 2.61) and SLEDAI-2K ≥4 reduction (OR 2.47) [2]. This apparent contradiction almost certainly reflects methodological differences in how each study handled population adjustment and which source trials were included, rather than a true difference in treatment effects. The one consistent cross-study finding on overall activity is that anifrolumab is associated with superior corticosteroid tapering capacity [1, 4], a finding supported by the anifrolumab-versus-placebo data from the TULIP trials.

**Cutaneous Manifestations**

| Outcome | Drug | Population | Effect Estimate | Time Point | Analysis Type |
|---|---|---|---|---|---|
| CLASI-A ≥50% reduction [8] | Anifrolumab vs. placebo | Baseline CLASI-A ≥10 (n=107 vs. 94) [8] | 46% vs. 25% [8] | Week 52 [8] | Post-hoc pooled [8] |
| CLASI-A ≥50% reduction [4] | Anifrolumab vs. placebo | Baseline CLASI-A ≥10, total population [4] | Difference 21.0 pp (95% CI 8.1–34.0); p=0.001 [4] | Week 12 [4] | Post-hoc pooled [4] |
| CLASI-A ≥50% reduction, IFNGS-high [4] | Anifrolumab vs. placebo | IFNGS-high, CLASI-A ≥10 [4] | Difference 22.6 pp (95% CI 8.4–36.9); p=0.002 [4] | Week 12 [4] | Post-hoc pooled [4] |
| CLASI-A ≥50% reduction, IFNGS-low [4] | Anifrolumab vs. placebo | IFNGS-low, CLASI-A ≥10 [4] | Difference 6.7 pp (95% CI −26.3 to 39.6); p=0.692 [4] | Week 12 [4] | Post-hoc pooled [4] |
| BILAG-2004 mucocutaneous response [8] | Anifrolumab vs. placebo | Pooled TULIP population [8] | 54% vs. 38% [8] | Week 52 [8] | Post-hoc pooled [8] |
| SLEDAI-2K mucocutaneous response [8] | Anifrolumab vs. placebo | Pooled TULIP population [8] | 55% vs. 39% [8] | Week 52 [8] | Post-hoc pooled [8] |
| CLASI response, overall [6] | Anifrolumab | Real-world CLE/SLE, n=13 [6] | 60% complete CLASI response; 100% overall response [6] | Mean 5 months [6] | Retrospective case series [6] |
| CLASI Activity, baseline to follow-up [5] | Anifrolumab | Refractory cutaneous SLE, n=18 [5] | Mean CLASI Activity 12.0 → 1.6 (SD 2.4) [5] | Mean 7.1 months [5] | Real-world case series [5] |
| Mucocutaneous response vs. placebo [3] | Both biologics | NMA, 11 trials [3] | Favored both drugs vs. placebo; no between-drug estimate [3] | Not reported [3] | NMA [3] |

The data on cutaneous involvement consistently and strongly favor anifrolumab. Among patients in the TULIP pooled analysis with baseline CLASI-A ≥10, anifrolumab produced a ≥50% CLASI-A reduction in 46% of patients at week 52 compared with 25% on placebo [8], and a 21-percentage-point treatment difference was established as early as week 12 [4]. This cutaneous benefit was most pronounced in IFNGS-high patients (22.6 pp difference) and absent in IFNGS-low patients (6.7 pp, non-significant) [4], indicating that anifrolumab's mechanism of type-I interferon receptor blockade is directly relevant to cutaneous disease activity in patients with an active interferon

signature. Real-world evidence reinforces this picture: in the Flouda series of 18 patients whose skin rash had failed an average of 4.5 prior immunosuppressive agents—including belimumab in 15 patients—anifrolumab reduced mean CLASI Activity from 12.0 to 1.6 over 7.1 months [5], and the Garbarino series reported a 100% overall response and 60% complete CLASI response across diverse CLE subtypes including chilblain lupus and panniculitis [6]. Notably, no significant efficacy differences were observed across CLE subtypes in that series [6], and response was similar in biologic-naive patients and those previously exposed to biologics [6]. The network meta-analysis confirmed that both biologics improve mucocutaneous outcomes relative to placebo, but provided no head-to-head estimate for this domain [3]. In sum, no controlled data demonstrate belimumab to be equivalent to anifrolumab for cutaneous disease, and real-world evidence consistently shows anifrolumab efficacy even after belimumab failure.

**Lupus Nephritis**

| Outcome | Drug | Population | Effect Estimate | Time Point | Analysis Type |
|---|---|---|---|---|---|
| Complete renal response (CRR), indirect comparison [1] | Belimumab vs. anifrolumab | LN patients in RCTs [1] | OR 0.83 (95% CI 0.22–3.10); not definitive [1] | Not reported [1] | Indirect meta-analysis [1] |
| Renal and cardiovascular/respiratory response [3] | Both biologics vs. placebo | NMA, 11 trials [3] | Not statistically significantly different from placebo for either drug [3] | Not reported [3] | NMA [3] |
| Complete renal response, real-world [9] | Belimumab | LN patients, n=10 [9] | 5/10 complete renal response at 1 year [9] | 12 months [9] | Retrospective observational [9] |
| Creatinine evolution [9] | Belimumab | LN patients, n=10 [9] | 0.713 ⟶ 0.763 ⟶ 0.783 mg/dL (baseline, 6, 12 months); P=0.748 [9] | 12 months [9] | Retrospective observational [9] |
| Proteinuria evolution [9] | Belimumab | LN patients, n=10 [9] | 0.822 ⟶ 1.027 ⟶ 0.796 g/24h (baseline, 6, 12 months); P=0.725 [9] | 12 months [9] | Retrospective observational [9] |
| Anti-dsDNA evolution [9] | Belimumab | LN patients, n=10 [9] | 162.5 ⟶ 74.33 ⟶ 72.00 U/mL (baseline, 6, 12 months); P=0.093 [9] | 12 months [9] | Retrospective observational [9] |
| LN class distribution (real-world belimumab) [10] | Belimumab | 9 patients with kidney involvement [10] | Class III 33.3%; class IV 22.2%; combined III+IV 22.2%; class V 11.1% [10] | — | Retrospective observational [10] |
| General renal/serological improvement [10] | Belimumab | LN subgroup, n=9 [10] | Qualitative improvement in serological activity, kidney function, and urine sediment [10] | Not reported [10] | Retrospective observational [10] |

In the domain of lupus nephritis, the evidence base is substantially weaker and the comparative picture is far less clear. The only indirect comparative estimate for complete renal response yielded OR 0.83 (95% CI 0.22–3.10), with a confidence interval so wide as to be clinically uninformative [1]. The network meta-analysis found no statistically significant renal benefit for either drug versus placebo [3]. Real-world belimumab data from two small cohorts document partial clinical improvements: the Rodríguez Tudero series observed complete renal response in 5 of 10 LN patients, with a decrease in anti-dsDNA (162.5 to 72.00 U/mL) without reaching statistical significance [9], while de la Rubia Navarro described qualitative improvement in renal function and serology in 9 patients [10]. Critically, anifrolumab is not approved for lupus nephritis and has been evaluated in only one published phase II trial in this indication [1], so no controlled anifrolumab data in LN patients are represented in this review. The real-world anifrolumab series were selected precisely for cutaneous disease and do not report renal outcomes [5, 6]. In this context, belimumab's established regulatory approval for LN represents a meaningful clinical distinction in the absence of anifrolumab LN data.

**Serological Activity**

| Outcome | Drug | Population | Effect Estimate | Time Point | Analysis Type |
|---|---|---|---|---|---|
| Anti-dsDNA normalization (indirect) [1] | Anifrolumab vs. belimumab | SLE RCTs [1] | OR 0.69 (95% CI 0.43–1.10); not statistically significant [1] | Not reported [1] | Indirect meta-analysis [1] |
| C3/C4 normalization (indirect) [1] | Anifrolumab vs. belimumab | SLE RCTs [1] | OR 0.69 (95% CI 0.42–1.13); not statistically significant [1] | Not reported [1] | Indirect meta-analysis [1] |
| BICLA response, abnormal serology subgroup [4] | Anifrolumab vs. placebo | ≥1 abnormal baseline serological marker [4] | Difference 23.1 pp (95% CI 14.1–32.1); p<0.001 [4] | Week 52 [4] | Post-hoc pooled [4] |
| BICLA response, normal serology subgroup [4] | Anifrolumab vs. placebo | Normal baseline serology [4] | Difference 8.5 pp (95% CI −2.7–19.7); p=0.137 [4] | Week 52 [4] | Post-hoc pooled [4] |
| Annualised flare rate, abnormal serology [4] | Anifrolumab vs. placebo | ≥1 abnormal baseline serological marker [4] | RR 0.61 (95% CI 0.48–0.77); p<0.001 [4] | 52 weeks [4] | Post-hoc pooled [4] |
| Annualised flare rate, normal serology [4] | Anifrolumab vs. placebo | Normal baseline serology [4] | RR 1.09 (95% CI 0.80–1.49); p=0.573 [4] | 52 weeks [4] | Post-hoc pooled [4] |
| SLEDAI-2K immunological domain improvement [8] | Anifrolumab vs. placebo | Pooled TULIP population [8] | 19% (44/237) vs. 11% (26/230) [8] | Week 52 [8] | Post-hoc pooled [8] |
| Anti-dsDNA evolution, real-world [9] | Belimumab | LN patients, n=10 [9] | 162.5 ⟶ 72.00 U/mL (baseline to 12 months); P=0.093 [9] | 12 months [9] | Retrospective observational [9] |
| Type-I IFN signature normalization [6] | Anifrolumab | CLE/SLE patients, n=6 assessed [6] | High z-scores normalized after treatment in all 6 [6] | Within months of treatment [6] | Retrospective case series [6] |

The serological activity domain reveals an important mechanistic distinction. Although neither indirect comparison of anti-dsDNA nor C3/C4 normalization reached statistical significance when comparing anifrolumab with belimumab [1], the TULIP post-hoc analyses show a clear serological moderator effect for anifrolumab: patients with at least one abnormal baseline serological marker achieved a 23.1 pp BICLA advantage over placebo (p<0.001), whereas the advantage in patients with normal serology was only 8.5 pp and non-significant (p=0.137) [4]. A parallel finding emerges for flare reduction, where the annualised flare rate ratio of 0.61 (p<0.001) with anifrolumab in serologically active patients compared with an RR of 1.09 (p=0.573) in serologically inactive patients indicates that anifrolumab's benefit is substantially concentrated in seropositive patients [4]. The type-I IFN signature normalization observed in all six assessed patients in the Garbarino series further supports the biological plausibility of preferential response in serologically active, high-interferon-signature patients [6].

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## Response Predictors

| Predictor | Drug | Outcome | Effect | Statistical Method |
|---|---|---|---|---|
| SLEDAI-2K ≥10 at baseline [7] | Belimumab | SRI-4 at 12 months [7] | OR 40.46; p=0.001 [7] | Multivariate logistic regression [7] |
| Polyarthritis at baseline [7] | Belimumab | SRI-4 at 12 months [7] | OR 12.64; p=0.001 [7] | Multivariate logistic regression [7] |
| SLEDAI-2K ≥10 at baseline [7] | Belimumab | SRI-4 at 24 months [7] | OR 15.97; p=0.008 [7] | Multivariate logistic regression [7] |
| Polyarthritis at baseline [7] | Belimumab | SRI-4 at 24 months [7] | OR 32.36; p=0.006 [7] | Multivariate logistic regression [7] |
| Prednisone ≥7.5 mg/day at baseline [7] | Belimumab | SRI-4 at 24 months [7] | OR 9.94; p=0.026 [7] | Multivariate logistic regression [7] |
| High IFNGS (IFNGS-high) [4] | Anifrolumab | BICLA at week 52 [4] | Difference 18.2 pp vs. 9.3 pp in IFNGS-low (non-significant) [4] | Stratified Cochran–Mantel–Haenszel analysis [4] |
| ≥1 abnormal baseline serological marker [4] | Anifrolumab | BICLA at week 52 [4] | Difference 23.1 pp vs. 8.5 pp with normal serology [4] | Stratified Cochran–Mantel–Haenszel analysis [4] |
| Active skin involvement (main indication) [5] | Anifrolumab | Cutaneous response [5] | Rapid early response in all patients; no formal predictor model [5] | None (observational case series) [5] |
| CLE subtype [6] | Anifrolumab | CLASI response [6] | No significant differences across subtypes [6] | None (retrospective case series) [6] |
| Prior biologic exposure [6] | Anifrolumab | Response [6] | No significant differences between biologic-naive and biologic-exposed [6] | None (retrospective case series) [6] |
| Factors associated with belimumab discontinuation [7] | Belimumab | Drug discontinuation [7] | None identified [7] | Multivariate logistic regression [7] |

The predictor data reveal divergent patient profiles that are associated with optimal response to each biologic. For belimumab, higher overall disease activity (SLEDAI-2K ≥10) and the presence of polyarthritis were the strongest independent predictors of SRI-4 response at both 12 and 24 months, with strikingly large odds ratios in the Iaccarino cohort [7]. This pattern—high disease activity and predominantly musculoskeletal involvement predicting

belimumab response—is consistent with the drug's mechanism of B-lymphocyte stimulator (BLyS) inhibition, which targets the systemic inflammatory and serological axis. For anifrolumab, the analogous predictors appear to be IFNGS-high status and baseline serological abnormality, with the largest treatment advantages over placebo observed in these subgroups [4]. The absence of any identified predictor of belimumab discontinuation in the Iaccarino study is noteworthy in that it suggests that drug survival in clinical practice is not readily anticipated from baseline characteristics [7]. Across real-world anifrolumab series, neither CLE subtype nor prior biologic exposure (including prior belimumab failure) moderated the response, implying that anifrolumab may retain utility even in heavily pre-treated cutaneous disease [6].

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## Safety and Treatment Discontinuation

| Safety Endpoint | Anifrolumab | Belimumab | Source | Notes |
|---|---|---|---|---|
| Serious adverse events (SAEs) [3] | Similar to belimumab [3] | Similar to anifrolumab [3] | NMA [3] | No rate reported [3] |
| SAEs, anifrolumab vs. placebo [4] | 11.1% [4] | — | TULIP pooled [4] | Placebo 16.4% [4] |
| Herpes zoster, NMA [3] | Higher vs. belimumab: OR 4.74 (95% CI 2.00–11.21) [3] | Reference [3] | NMA [3] | Statistically significant [3] |
| Herpes zoster, indirect meta-analysis [1] | 7.2% [1] | 3.2–4.6% [1] | Indirect meta-analysis [1] | — |
| Herpes zoster, TULIP pooled [4] | 6.4% [4] | — | TULIP pooled [4] | Placebo 1.4% [4] |
| Herpes zoster, real-world anifrolumab [5] | 1/18 (discontinued) [5] | — | Real-world case series [5] | Led to discontinuation [5] |
| Herpes zoster, real-world anifrolumab (Garbarino) [6] | 0/13 [6] | — | Real-world case series [6] | All patients vaccinated prior to starting [6] |
| Serious infections, NMA [3] | Similar to belimumab [3] | Similar to anifrolumab [3] | NMA [3] | No rate reported [3] |
| Non-opportunistic serious infections, TULIP [4] | 4.4% [4] | — | TULIP pooled [4] | Placebo 6.0% [4] |
| Severe adverse events, belimumab real-world [7] | — | Low rate; not quantified [7] | Italian prospective cohorts [7] | — |
| Belimumab discontinuation rate [7] | — | 30.8% (58/188) after mean 10.4 months [7] | Italian prospective cohorts [7] | No predictors of discontinuation identified [7] |
| Belimumab drug survival [7] | — | 86.9% (6 mo); 76.9% (12 mo); 69.4% (18 mo); 67.1% (24 mo); 61.9% (30 mo) [7] | Italian prospective cohorts [7] | — |

| Safety Endpoint | Anifrolumab | Belimumab | Source | Notes |
|---|---|---|---|---|
| Anifrolumab discontinuation, real-world [5] | 2/18 at 4 and 3 months [5] | — | Real-world case series [5] | Reasons: zoster infection; generalized lymphadenopathy [5] |
| Belimumab-to-anifrolumab switch [9] | — | 1 patient interrupted for lupus psychosis [9] | Real-world observational [9] | 1 patient switched to anifrolumab for refractory skin despite renal response [9] |

The safety comparison is the one domain where there is substantial consistency across heterogeneous study designs. In both the network meta-analysis and the indirect meta-analysis, SAEs and serious infections were found to be broadly comparable between the two biologics [1, 3], and SAE rates in the TULIP pooled analysis showed anifrolumab at 11.1% versus placebo at 16.4% [4]. The single consistently replicated safety signal is the elevated rate of herpes zoster with anifrolumab. The NMA yielded an OR of 4.74 (95% CI 2.00–11.21) for herpes zoster with anifrolumab versus belimumab [3]; the indirect meta-analysis reported rates of 7.2% for anifrolumab versus 3.2–4.6% for belimumab [1]; and the TULIP pooled data show a 6.4% rate with anifrolumab compared with 1.4% with placebo [4]. In real-world practice, zoster infection caused discontinuation in one of 18 patients in the Flouda series [5], while the Garbarino series recorded no zoster cases—likely reflecting their practice of completing the vaccination schedule before initiating treatment [6]. This observation is clinically important: prior vaccination may substantially mitigate anifrolumab's principal excess safety risk. For belimumab, the Iaccarino cohort documented 30.8% discontinuation overall over a mean of 10.4 months, with drug survival falling to 61.9% at 30 months [7], though no baseline predictors of discontinuation were identified [7].

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## Synthesis

### Apparent Contradictions in SRI-4 Comparative Efficacy

The most prominent inconsistency across sources concerns which drug achieves greater SRI-4 response. Lee & Song concluded that belimumab was superior (OR 1.99, CI 1.25–3.16) [1], while Bruce et al. reached the opposite conclusion, reporting anifrolumab superior (OR 2.61, CI 1.22–5.58) [2], and the Martínez-Martínez NMA found no significant difference between the two [3]. These three analyses used different methodological approaches to the same absence of head-to-head data: NMA, indirect meta-analysis, and population-adjusted simulated treatment comparison respectively. The Bruce et al. analysis explicitly used individual patient data from TULIP-1/2 to adjust for population imbalances against summary-level data from BLISS-52/76 [2], which represents a more sophisticated approach to the fundamental problem of cross-trial population differences than an unadjusted indirect comparison. However, even population-adjusted indirect comparisons remain sensitive to unobserved confounding driven by inclusion criteria differences between the TULIP and BLISS trial programmes. The most appropriate interpretation of these three discordant findings is that SRI-4 equivalence between the two drugs cannot be excluded, and that the directional advantage observed in either indirect analysis likely reflects residual population heterogeneity between the TULIP and BLISS trial populations rather than a true differential pharmacological effect on this composite outcome.

### Context-Specific Conclusions by Clinical Profile

Notwithstanding the ambiguity around overall SRI-4 response, the evidence converges on clinically meaningful profile-specific conclusions when the focus shifts from composite activity scores to domain-specific outcomes.

For cutaneous disease, the evidence consistently and substantially favors anifrolumab. The controlled trial data from the TULIP programme, supported by two real-world series encompassing a combined 31 patients with predominantly refractory skin disease, all point in the same direction: large, early reductions in CLASI-A that are independent of CLE

subtype and that persist even after multiple prior biologic failures including belimumab [4–6, 8]. The mechanistic basis is clear: type-I interferon signalling is strongly implicated in the pathogenesis of cutaneous lupus, and IFNGS-high status amplifies the CLASI-A response advantage [4]. The practical implication is that patients with prominent or refractory cutaneous manifestations, particularly those with an elevated type-I IFN signature, are the group for whom anifrolumab's advantage over belimumab is best supported.

For lupus nephritis, the reverse situation applies: belimumab has regulatory approval for LN supported by controlled trial data, while anifrolumab has not been studied in this indication beyond a single phase II trial [1]. The indirect CRR comparison was statistically uninformative (OR 0.83, CI 0.22–3.10) [1], and the NMA found no significant renal benefit for either drug versus placebo [3]. Real-world belimumab data in small LN cohorts document partial clinical responses in renal parameters, with the anti-dsDNA trajectory in the Rodríguez Tudero series suggesting biological activity even without statistically significant changes in proteinuria or creatinine [9]. Until controlled anifrolumab data in LN are available, the comparative efficacy question in this domain is empirically unanswerable.

For patients with high overall disease activity and musculoskeletal involvement, the Iaccarino predictor analysis provides the most direct clinical guidance: baseline SLEDAI-2K ≥10 and polyarthritis were the strongest independent predictors of belimumab SRI-4 response, with odds ratios of 40.46 and 12.64 at 12 months respectively [7]. No equivalent validated predictor model exists for anifrolumab. While IFNGS-high status and serological abnormality identify subgroups with a larger BICLA advantage with anifrolumab versus placebo [4], these have not been validated in independent cohorts as prospective selection criteria.

For serologically active SLE without a dominant single-organ phenotype, both mechanistic reasoning and the TULIP subgroup data suggest that anifrolumab's benefit is particularly concentrated here. The difference in BICLA response between anifrolumab and placebo was 23.1 pp in patients with abnormal serology but only 8.5 pp (non-significant) in those with normal serology [4], and flare reduction was similarly restricted to the seropositive group [4]. This pattern suggests that patients with normal baseline serology derive little measurable benefit from anifrolumab, whereas the converse—high serological activity as a predictor of belimumab response—is less clearly established from the available data.

The herpes zoster risk differential (NMA OR 4.74 for anifrolumab vs. belimumab) [3] must enter any individualized treatment decision. The Garbarino series observed no zoster cases among patients who completed pre-treatment vaccination [6], which suggests that vaccination status should be a routine prerequisite rather than a contraindication, and that the excess risk may be attenuated in adequately vaccinated populations. What remains unresolved is whether there are patient subgroups—for example, patients with prior zoster history or intensive concomitant immunosuppression—for whom the zoster risk with anifrolumab is disproportionately elevated and for whom belimumab would be the preferred biologic regardless of cutaneous disease burden.

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